THOMPSON CHIDOZIE ONAH

thompson.onah.247023@unn.edu.ng | +234 810 790 0605 | Anambra State, Nigeria

PROFESSIONAL SUMMARY

Electronic Engineering graduate with hands-on experience in embedded systems, sensor interfacing and power electronics, gained through a closed-loop control thesis project and industrial training in solar installation and inverter repair. Seeking graduate research opportunities.

RESEARCH INTEREST

Embedded systems . sensing and signal conditioning . power electronics . control system · RF and microwave engineering · electromagnetic wave engineering for wireless communications

EDUCATION

University of Nigeria, Nsukka (UNN) *Aug 2021 - Sep 2025*

B.Eng. in Electronic Engineering

- Graduated with Second Class Honours, Upper Division; Overall CGPA: **3.67/5.0** (final two years): **4.46 / 5.0**
- Undergraduate Thesis: "Automatic AC Fan Regulation System Using Temperature Sensing" (Grade: A)
- Relevant coursework: Control Theory, Digital Signal Processing, Micro processing Systems, Computer System Design, Communication Systems, Reliability Engineering

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher - University of Nigeria, Nsukka *Mar 2025 - Nov 2025*

Automatic AC Fan Regulation System Using Temperature Sensing

- Designed and implemented a closed-loop, temperature-sensing control system that automatically regulates AC fan speed to improve indoor thermal comfort and energy efficiency. Implemented using an atmega8 micro controller and a TRIAC module to achieve precise fan speed control. With a dht11 temperature sensor
- Built and tested the sensing, signal-conditioning, and control circuitry, then validated system performance through bench experiments. The overall response time was measured at about 50–60 milliseconds, and accuracy in matching fan speed to temperature thresholds was recorded at 98.79 percent.
- Documented the design methodology and experimental results in a thesis.

PROFESSIONAL EXPERIENCE

Fouani Nigeria Limited - Solar Installation Department *Aug 2024 - Oct 2024*

- Installed and commissioned residential and commercial solar PV systems, including hybrid inverters and lithium battery storage.
- Carried out DC/AC wiring, system testing, and commissioning to ensure safe, reliable operation.

Fouani Nigeria Limited - Solar Service Centre *Nov 2024 - Jan 2025*

- Diagnosed and repaired faults in inverters and solar power systems.
- Provided preventive maintenance and technical support to customers.

TECHNICAL SKILLS

Programming & Simulation: C++, Python, MATLAB, VHDL, Arduino, Dart, SPICE-level circuit simulation

Embedded Systems: Arduino, microcontrollers, sensor integration & real-time data acquisition (I2C LCD, Hall-effect sensors, BMP180, DHT11, RTC1307, SD module)

Additional project: Affordify — mobile application built in Dart/Flutter, published on GitHub.

PUBLICATIONS, AFFILIATIONS & LINKS

Council for the Regulation of Engineering in Nigeria (COREN) *July 2026 - Present*

- Registered Pupil Engineer (P. Engr./P-COREN)

Nigerian Society of Engineers (NSE) *July 2026 - Present*

- Graduate Member (GMNSE)

Onah, T. C., Ogara, D. N**.,** & Onah, V. O. "Automatic AC Fan Regulation System Using Temperature Sensing." *IECEC 2026*, University of Nigeria, Nsukka. Supervisor: Engr. Dr. C. M. Ogbuka.

Role: Co-author; contributed to the design, hardware implementation, temperature sensing, testing, and documentation of the system.

GITHUB